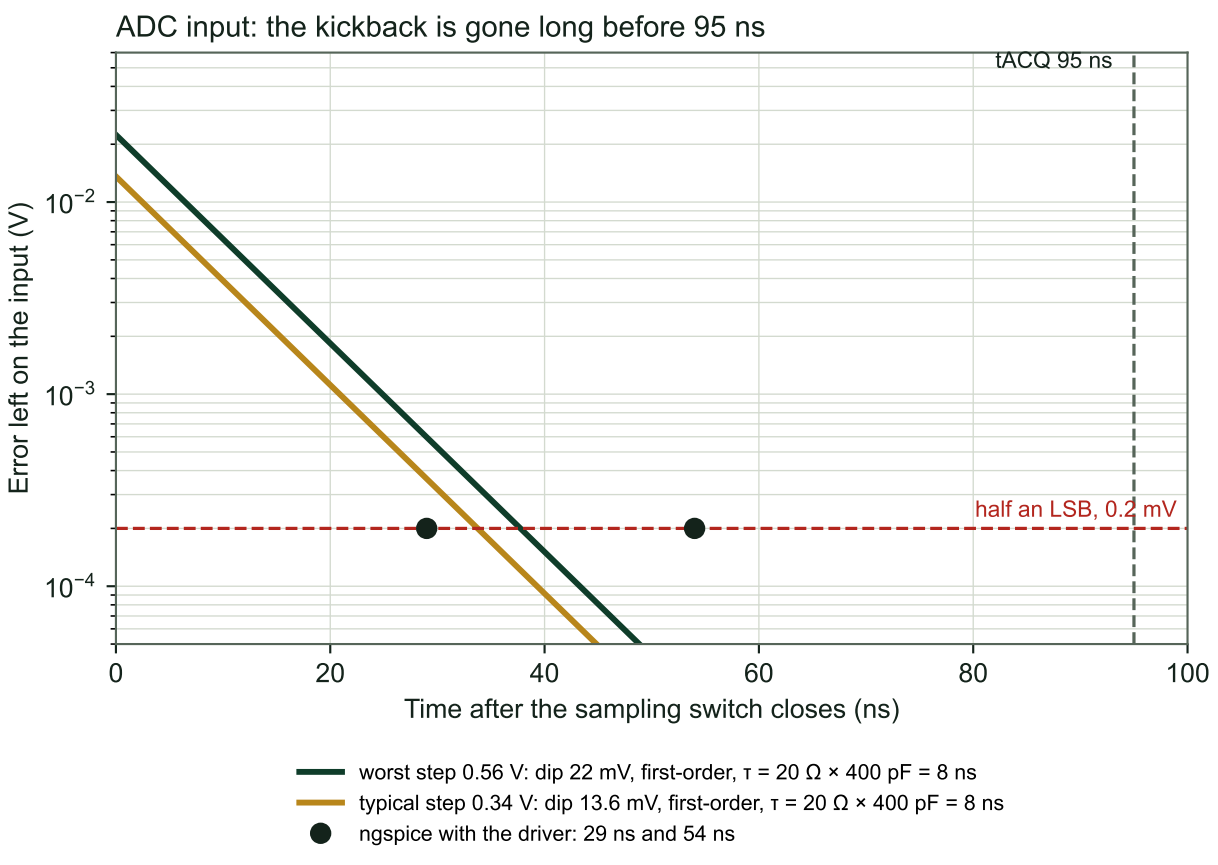
ADC input: the kickback is gone long before 95 ns
tACQ 95 ns
half an LSB, 0.2 mV
Error left on the input (V)
10^{-2}
10^{-3}
10^{-4}
0
20
40
60
80
100
Time after the sampling switch closes (ns)
worst step 0.56 V: dip 22 mV, first-order, τ = 20 Ω × 400 pF = 8 ns
typical step 0.34 V: dip 13.6 mV, first-order, τ = 20 Ω × 400 pF = 8 ns
ngspice with the driver: 29 ns and 54 ns